

Project ID :

25-26J-207

1. Topic (12 words max)

Inclusive Mobile Learning App for Visually, Hearing, Physically and Cognitively Impaired Students (Grade 3-5)

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Many digital learning tools today are not fully inclusive for students with different types of disabilities. Most existing educational apps focus on only one disability type-such as hearing or learning difficulties-and often cover just basic lessons like letters or numbers. For example, the Visual Kids app in Sri Lanka is made only for hearing-impaired children and focuses on early literacy, but it does not support students with other disabilities like blindness or physical challenges.

Some research has introduced helpful tools like screen readers or sign language videos, but these tools are often used separately and not in one platform. Many global apps also do not focus on the needs of Sri Lankan students, their local language, or their specific curriculum. Other solutions like MoodleAcc+ follow international accessibility rules but are designed for general users, not for young students with special needs. Also, many of these tools are not fun or interactive enough to keep children engaged.

There is also a lack of adaptive interfaces that change based on each student's abilities. Students with physical disabilities may struggle with small buttons, while those with cognitive challenges need clear instructions, friendly visuals, and simple layouts. Current systems do not provide these features in one easy-to-use app. Furthermore, most apps don't offer fun learning with real-time feedback or quizzes tailored to different disability types.

The main research problem is that there is no single learning platform that supports multiple types of disabilities, matches Sri Lankan educational needs, and uses adaptive technologies to help children learn in a fun, personalized, and accessible way. Students with visual, hearing, physical, or cognitive difficulties still face many barriers when trying to access digital learning, especially in local school settings. This project aims to solve that problem by creating an all-in-one mobile app that provides personalized learning tools, content, and interfaces for children with different abilities in Grades 3 to 5. This will support fair and effective learning for every child, no matter their disability.

References

- [1] J. M. D. G. K. M. Dodandeniya, G. A. I. P. Dayananda, S. N. N. Hewagama, W. W. M. R. T. I. W. Dela, D. R. Wijendra, and S. Kasthuriarachchi, "Visual Kids: Interactive Learning Application for Hearing-Impaired Primary School Kids in Sri Lanka," in *Proc. 5th Int. Conf. on Advancements in Computing (ICAC)*, Colombo, Sri Lanka, Dec. 2023, pp. 846–851. doi: [10.1109/ICAC60630.2023.10417450](https://doi.org/10.1109/ICAC60630.2023.10417450)
- [2] K. R. A. Kumar, S. Ravi, and S. K. Srivatsa, "Effective E-learning Approach for Students with Learning Disabilities," *Int. J. of Scientific & Engineering Research*, vol. 2, no. 11, pp. 1–5, Nov. 2011. [Online]. Available: <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=f74de60cdbeb5ee260f11b9d534ab0cf7a9811a1>
- [3] V. Bravou and A. Drigas, "A Contemporary View on Online and Web Tools for Students with Sensory & Learning Disabilities," *Int. J. of Online and Biomedical Engineering (iJOE)*, vol. 15, no. 12, pp. 97–105, 2019. doi: [10.3991/ijoe.v15i12.10833](https://doi.org/10.3991/ijoe.v15i12.10833)
- [4] M. Laabidi, M. Jemni, L. Jemni Ben Ayed, H. Ben Brahim, and A. Ben Jemaa, "Learning technologies for people with disabilities," *J. of King Saud Univ. – Comput. and Inf. Sci.*, vol. 26, no. 1, pp. 29–45, 2014. doi: [10.1016/j.jksuci.2013.10.005](https://doi.org/10.1016/j.jksuci.2013.10.005)
- [5] M. Shonfeld and I. Ronen, "Online Learning for Students from Diverse Backgrounds: Learning Disability Students, Excellent Students and Average Students," *IAFOR J. of Education*, vol. 3, no. 2, pp. 13–29, 2015. [Online]. Available: <https://files.eric.ed.gov/fulltext/EJ1100564.pdf>

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This project introduces a mobile learning app for Sri Lankan primary students (Grades 3–5) with disabilities, including visual, hearing, physical, and cognitive challenges. The goal is to provide one simple platform that adjusts to each child's specific needs and ensures equal access to learning.

At registration, the student (or a guardian/teacher) selects the disability type. Based on this input, the app automatically updates its interface and features accordingly:

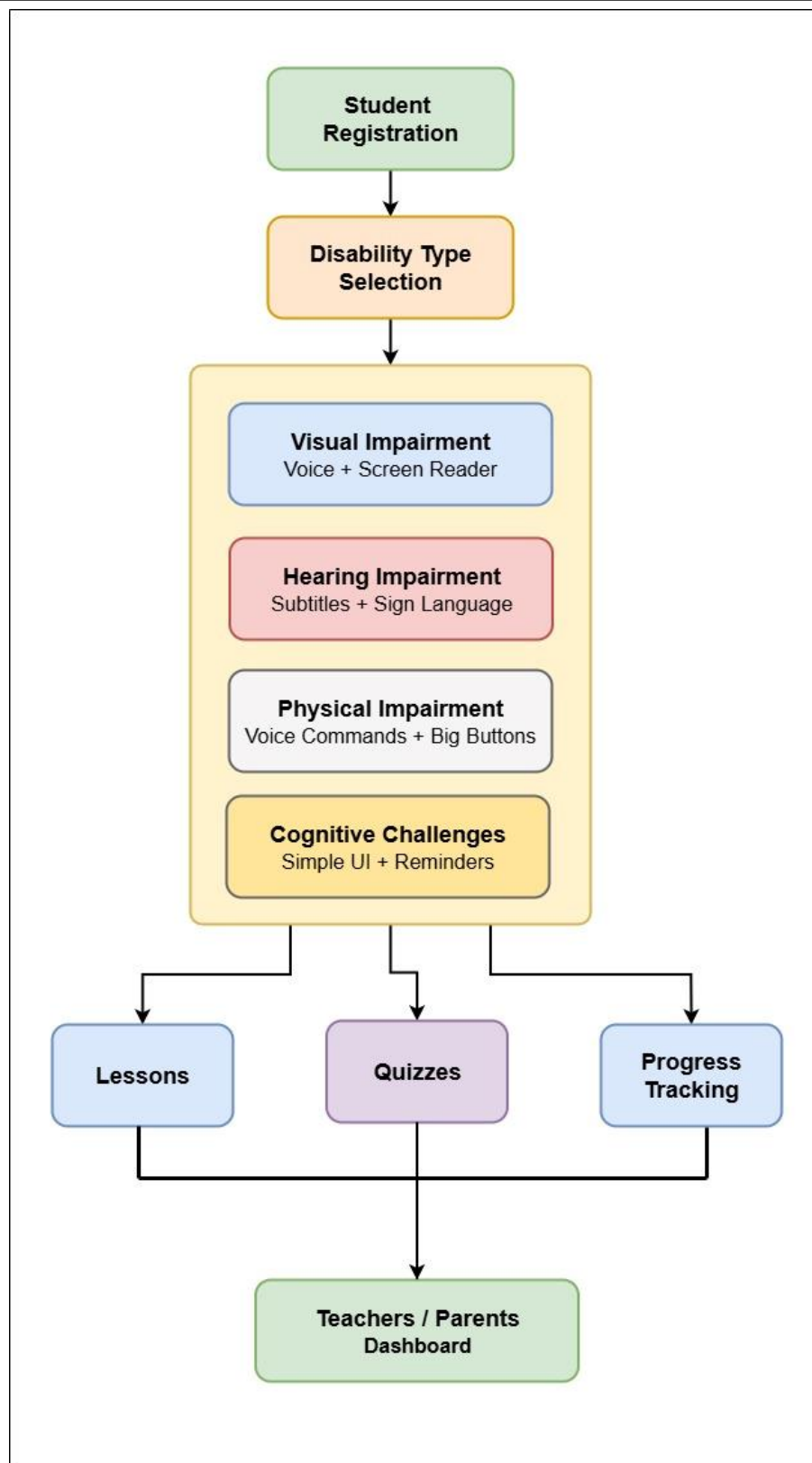
- Visually impaired students receive voice instructions and screen reader support.
- Hearing-impaired students are shown subtitles, transcripts, and sign language videos.
- Physically challenged students can use voice commands, large buttons, or keyboard navigation.
- Students with cognitive challenges get step-by-step guidance, simplified visuals, and friendly reminders.

The app is developed using Flutter, enabling cross-platform deployment with built-in accessibility services such as semantic widgets, focus management, responsive UI, and integration with device sensors (microphone, camera, gyroscope). It follows WCAG 2.1 Level AA guidelines, ensuring the interface is perceivable, operable, understandable, and robust for all users.

Core features include:

- AI-generated quizzes to test understanding
- Progress tracking for parents and teachers
- Tools such as text-to-speech, speech-to-text, subtitle support, and interactive avatars

This solution aims to reduce the digital education gap by delivering a fair, inclusive, and engaging learning experience for every child with special needs in Sri Lanka.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research project focuses on developing an inclusive mobile learning platform for students with disabilities, especially from Grade 3 to Grade 5. To develop a meaningful and effective system, we need specialized knowledge, domain expertise, and real data related to these students and their learning environments.

Domain Expertise:

We will consult with:

- Special education teachers to understand how each type of disabled student learns.
- Speech and language therapists, sign language experts, and occupational therapists to design suitable learning tools and assistive features.
- School administrators to understand current challenges in inclusive learning environments.

Data Requirements:

To design and test the app correctly, we plan to collect real data from special units in government schools that support students with disabilities. This includes:

- Government special schools and Navajeevana Rehabilitation School in Tangalle.
- Details on types and numbers of students with disabilities.
- Syllabus, textbooks, audio-visual materials, and tools currently used.
- Assistive devices used in classrooms (e.g., screen readers, hearing aids).
- Observations from real classroom settings and interviews with teachers and parents.
- Preferred language (Sinhala) and digital literacy of students and facilitators.

All data will be handled confidentially and used only for academic research. Consent will be obtained from schools and guardians before collection.

The platform will be developed using Flutter, which allows cross-platform compatibility and smooth integration with mobile device sensors such as the microphone, gyroscope, and camera. The system will be designed according to WCAG 2.1 Level AA accessibility standards to ensure full usability for students with visual, hearing, physical, and cognitive impairments. By combining real field data with expert input and modern mobile technologies, this research aims to build a truly inclusive and locally relevant solution for differently-abled learners in Sri Lanka.

8. Objectives and Novelty

Main Objective To develop a mobile learning app for Grade 3–5 students with disabilities that supports Sinhala language and provides accessible, voice-based, and easy-to-use learning tools for visual, hearing, physical, and cognitive challenges.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Wijesooriya C.D. IT22078728	Visual Impairment Support	- Enables voice-based interaction for navigation, lesson flow, and quizzes. - Integrates Text-to-Speech (TTS) and Speech-to-Text (STT) via Flutter plugins. - Uses stereo spatial audio cues to simulate learning environments. Ex: In a jungle lesson, bird sounds come from the left and waterfall sounds from behind to help students imagine their surroundings. - Includes biometric login using Face ID. - Compatible with screen readers like TalkBack and VoiceOver. - Allows generation of Braille-ready learning documents using an AI assistant.	- AI-powered voice assistant for guided instruction. - Vibration feedback for actions. - Contextual auditory prompts for real-time guidance. - AI-supported Braille document generator for tactile learning. 3D directional sound-based storytelling for immersive learning.

Thanthirige K.S.G.I. IT22177650	Hearing Impairment Support	• Lessons use Sri Lankan Sign Language (SLSL) in video format. • Camera-based sign recognition via pose estimation. • Uses chatbot and visual instructions with local language support. • Quizzes and feedback are fully text-based.	• Gesture scoring using camera pose detection. • Sign imitation games. • Animated visual explanations in local language.
Perera G.W.R.S. IT22064622	Physical Impairment Support	• Supports one-switch and voice-controlled navigation. • Auto-play for lessons with minimal interaction. • Optimized for large buttons. • Integrates Bluetooth assistive devices (e.g., switches). • Adds dwell-time activation to buttons for hands-free control (e.g., hold toe/elbow gently). • Visual countdown indicators for safe button selection. • Allows cancellation of accidental touches by lifting before timer ends.	• Gaze tracking for control. • Tilt-based navigation using the device accelerometer. • Facial gesture input for hands-free access. • Dwell-time activation for users with no finger control – touch and hold with any body part. • Voice or sound-based emergency “Stop” gesture (say "Stop" or hum). • Customizable dwell time, feedback types, and button size to fit student needs.
Mathota Arachchi S.S IT22070876	Cognitive & Learning Difficulties Support	• Uses cartoon characters, storytelling, and drag-drop games for interaction. • Provides short, emotion-based feedback cycles.	• Avatar-based emotional modulation. • Adaptive learning paths tailored to learning speed.

		• Adapts flow based on attention span and interaction. • Lessons presented with simple visuals, one instruction at a time.	• Reinforcement schedules for cognitive engagement.
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9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Wijesooriya C.D. IT22078728	Designed for students who are blind or have low vision. The module provides full voice-based interaction using Text-to-Speech (TTS) and Speech-to-Text (STT), allowing students to navigate, learn, and take quizzes without needing to look at the screen. It also integrates with screen readers (TalkBack / VoiceOver), provides spatial audio cues for better understanding, and includes Face ID for secure login.
Thanthirige K.S.G.I. IT22177650	Tailored for students who are deaf or hard of hearing. All lessons are presented through text and Sri Lankan Sign Language (SLSL) videos. A camera-based gesture recognition system helps students practice signs. The app also offers a chatbot and animated visual content to help students understand concepts without relying on sound.
Perera G.W.R.S. IT22064622	Built for students with motor impairments who struggle with physical movement. The app supports voice commands, large tap areas, and one-switch access. It also integrates Bluetooth assistive devices and plans to support gaze tracking and tilt-based navigation, allowing complete control with minimal effort.
Mathota Arachchi S.S IT22070876	Developed for students with cognitive or learning difficulties such as trouble with attention, memory, or understanding. The module uses short instructions, cartoon characters, and drag-drop games to engage students. Lessons are broken down into small steps, with visual storytelling, friendly feedback, and adaptive learning based on user behavior.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Prof.	Nuwan	Kodagoda	
Co-Supervisor	Mrs.	Jenny	Krishna	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.